

Constraints

- Key Constraint (primary key vs unique)
- Referential Integrity Constraint (FK)

$$R \subseteq S \Rightarrow R - S = \emptyset$$

$$R = \emptyset \Rightarrow R \subseteq \emptyset$$

SQL Commands

Foreign Keys

```
CREATE TABLE Studio (  
  name CHAR(30) PRIMARY KEY,  
  address VARCHAR(255),  
  presC INT,  
  FOREIGN KEY (presC) REFERENCES MovieExec (cert)  
);
```

Insert

```
INSERT INTO  
  Movies  
VALUES  
  (  
    "Screaming",  
    Date '2023-10-31',  
    "Adventure",  
    "CSUC"  
  );
```

AS

```
SELECT  
  title AS name  
FROM  
  movies;  
  
SELECT  
  title AS name,  
  length * 0.0167 AS lengthInHours;
```

LIKE

-% is regex *

- _ is regex .

NULL

- Arithmetic on NULL results in NULL
- Comparison on NULL results in UNKNOWN

Check if NULL

```
X IS NULL;  
X IS NOT NULL;
```

ORDER BY

```
SELECT  
  *  
FROM
```

```
    Movies
WHERE
    YEAR = 1990
ORDER BY
    length,
    title;

SELECT
    *
FROM
    R
ORDER BY
    A + B DESC;
```

Good Design

- Faithfulness